# Falcon CMS

## Scenario Test Report

CMS Core V2 API verification mapped to Falcon CMS API and Events v2 Table 3

| Item | Value |
|---|---|
| Report date | 21 October 2026 |
| Report version | 2.3 |
| Scenario source | Falcon CMS – API & Events v2 (Table 3, 42 scenarios); SHA-256 fb0ad93bb03edb5f… |
| Assessment basis | Consolidated CMS evidence through 21 October 2026. Both complete 20 October reruns used WAIT_FOR_SWITCH_SYNC=true; focused W15 and W23 results are included. See section 3 for run-specific settings and outcomes. |
| Test plan | CMS_V2_LIFECYCLE_FUNCTIONAL.jmx (JMeter 5.6.3), 1076 samplers, SHA-256 f28a6590f5a18eca… |
| Companion data | Baseline request/response evidence plus dated rerun files and current assessment under results/ (Appendices A and B) |
| Prepared by | CMS Core V2 engineering and QA |
| Classification | Confidential – prepared for the Falcon programme |

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# 1 Summary

The consolidated assessment through 21 October 2026 covers all 25 CMS scenarios in Table 3 phases 1 to 5 and 8 to 10: 20 PASS, 5 PARTIAL and 0 FAIL within the recorded CMS scope. Both complete 20 October reruns enabled Switch waits. Later focused runs verified W15 limits and corrected W23 key rotation. These scenario assessments combine evidence from several runs; they do not mean every sample in the latest complete run passed.

The remaining gaps are final authorisation on the CMS test cards, successful PIN operations, tenant isolation, and unresolved Switch wait timeouts recorded in section 3. W23 activation and post-rotation issuance passed on 21 October; F-01 is closed.

Standalone Switch transaction evidence is provided in Falcon_Switch_Test_Report.pdf. CMS scenario results, integration checks and execution totals are retained in this report.

## 1.1 Scenario outcome

| ID | Scenario | Passed / executed | Status | Note |
|---|---|---|---|---|
| W01 | Create and configure an institution | 29/29 | PASS | |
| W02 | Prepare HSM and institution data keys | 19/19 | PASS | |
| W03 | Configure BIN and BIN range | 36/36 | PASS | |
| W04 | Configure HSM keys for the BIN | 18/18 | PASS | PVK and ZPK assignments confirmed active |
| W05 | Create a product and its first version | 7/7 | PASS | |
| W06 | Configure product policies and limits | 36/36 | PASS | |
| W07 | Configure design, technology and CVM | 21/21 | PASS | |
| W08 | Link the range and design to product, then activate | 24/24 | PASS | |
| W09 | Issue a virtual-only card | 36/36 | PASS | |
| W10 | Issue a physical-only card | 18/18 | PASS | |
| W11 | Issue virtual first, then request physical | 21/21 | PASS | |
| W12 | Issue virtual and physical together | 5/5 | PASS | |
| W13 | Issue a physical card with multiple virtual PAN capability | 14/14 | PASS | |
| W14 | Change card controls | 27/27 | PASS | Switch acknowledgement confirmed on 20 October at revision 7 |
| W15 | Change per-transaction limits | 9/9 (+1 n/e) | PARTIAL | Known test-bed limitation: simulator uses a different card; limit authorisation unverified |
| W16 | Change and reset PIN | 14/14 | PARTIAL | Known test-bed limitation: different simulator card; positive PIN operations also untested |
| W17 | Hot-list a card | 12/12 | PASS | Operations and cardholder hot-list confirmed |
| W18 | Replace a lost, stolen or compromised card | 30/30 | PASS | |
| W19 | Replace a damaged or expired card | 44/44 | PARTIAL | Known test-bed limitation: simulator uses a different card; old-card authorisation unverified |
| W20 | Launch Product Version 2 with changed defaults | 41/41 | PASS | |

| ID | Scenario | Passed / executed | Status | Note |
|---|---|---|---|---|
| W21 | Launch a new card-design version | 11/11 | PASS | |
| W22 | Expand the BIN range available to a product | 12/12 | PASS | |
| W23 | Rotate an HSM key used by a BIN | 11/11 | PARTIAL | F-01 closed; known simulator limitation leaves old-card authorisation unverified |
| W24 | Run multiple products on one BIN | 13/13 | PASS | |
| W25 | Run the complete lifecycle smoke test | 57/57 baseline | PARTIAL | Combined evidence; final authorisation unverified and PIN coverage limited to validation |

Counts reflect the displayed baseline checks and dated additions. W25 groups 57 baseline checks into lifecycle legs and includes later hot-list evidence. Counts overlap across scenarios and must not be added as a unique suite total. PARTIAL identifies the stated coverage gaps. The NPCI simulator uses a different transaction card; successful PIN selection, change and reset are also unverified in W16 and the W25 PIN leg.

# 2 Scope and approach

## 2.1 Scope

- The report combines the 16 October CMS baseline with the dated verification runs in section 3. The original baseline samples and totals remain archived; scenario tables identify later evidence that supplements or supersedes them.
- W26 to W43 transaction evidence, including the latest NPCI simulator run and selected older examples, is provided separately in Falcon_Switch_Test_Report.pdf. Full accounting and CVM requirements remain unverified.
- Both complete 20 October reruns used WAIT_FOR_SWITCH_SYNC=true. W15 limit boundaries were subsequently tested with the local Accounting stub. Integration with live Accounting and final transaction authorisation remain unverified.

## 2.2 Method

- The packaged JMeter plan is the 16 October baseline plan (1,076 samplers; hash shown on the cover). Later reruns added or selected checks, and the 21 October W15/W23 checks used focused API scripts. Their scope and sources are identified separately.
- Every HTTP call carries an explicit assertion on status code and response content (Groovy/JSON assertions). Asynchronous operations (issuance, replacement) are polled and then required to reach their terminal state.
- Negative checks assert the controlled error and then read the resource back to prove that nothing changed.
- The JMeter functional run uses test data isolated by a unique run suffix.
- The companion JSON retains baseline requests, responses and assertions with sensitive data redacted. The dated rerun JSON files contain sanitised verification summaries; they do not provide full request/response payloads for every later check.

## 2.3 Environment

| Component | Detail |
|---|---|
| CMS Core V2 | Spring Boot service, Java 25, port 9800; PostgreSQL 18 (schema cms_v2) |
| Customer service | Customer, account and address fixtures via REST, port 9801 |
| Embossing service | Personalisation batch and file reads, port 9803 |
| Identity | Keycloak realm cms, OAuth2 client-credentials (platform-scoped test client) |

| Component | Detail |
|---|---|
| Switch | Falcon Switch core 1.0.58, port 8090, in the 20 October reruns. Both used Switch waits; successes and timeouts are recorded separately in sections 3.2 and 3.3. Baseline 16 October used 1.0.52. |
| HSM | Configured HSM provider; health verified at run start (00.4) |
| Test tooling | JMeter 5.6.3 on Java 21 for the historical suite runs; focused API scripts and seven Java regression tests for the 21 October W23 correction |

## 2.4 Result conventions

| Term | Meaning |
|---|---|
| PASS | The call returned the expected status and every assertion on the response and on the read-back held. |
| FAIL | A required assertion or wait did not succeed. The dated run evidence and associated scenario or integration note describe the failure. |
| PARTIAL (scenario) | The displayed CMS checks passed within their stated scope, but a required scenario leg or coverage item remains unverified. This is a consolidated scenario assessment, not a per-run sample status. |
| Not executed | The check was skipped by design because a prerequisite was not met in this run (for example a prior defect or a disabled dependency). |
| Check ID | S-nnnn identifies a baseline sample. Dated overrides use the indicated rerun file. R04, R14 and R17 references identify added report rows mapped to plan steps in results/report-current-assessment-2026-10-21.json. |

# 3 Execution summary

| Item | Value |
|---|---|
| Assessment cutoff | 21 October 2026; consolidated evidence from the baseline and dated reruns |
| Scenario assessment | 20 PASS / 5 PARTIAL / 0 FAIL within recorded CMS scope |
| Switch waits in complete 20 October runs | WAIT_FOR_SWITCH_SYNC=true in both the 06:24 Switch-sync run and the 07:37 PVK/ZPK run |
| Successful acknowledgement evidence | 20 October 06:24:15-06:32:21 UTC: 1,223 samples; controls, registration, activation, freeze/release and separate generation profiles confirmed. A closed-card wait timed out (section 3.2). |
| Later complete rerun | 20 October 07:37:05-07:48:57 UTC: 1,314 samples. PVK/ZPK checks passed; four Switch waits timed out. Seven failing samples were recorded (section 3.3). |
| Focused 21 October verification | W15: stub-backed limits and Switch profile revision 7 confirmed. W23: activation, assignment readback, issuance and historical key reference passed; seven regression tests passed. |
| Historical 16 October baseline only | FALCON-CMS-FUNC-2026-10-16; 07:17:12.550-07:21:10.093 UTC. 1,097 samples; 582 checks: 575 passed / 2 failed / 5 not executed. WAIT_FOR_SWITCH_SYNC=false applies only to this archived baseline. |
| Open integration and coverage items | Later Switch timeouts are not evidenced as resolved. Positive PIN operations, tenant isolation, live Accounting integration and authorisation on CMS test cards remain unverified. |

## 3.1 Historical baseline fixtures from 16 October 2026

| Object | Identifiers |
|---|---|
| Institution | institutionId: 01a0a5f5-f71d-7d0f-ba69-a020a51bbbc1, institutionCode: FLB_MVAMXDWE, origin: Existing CMS institution mapped to the Switch test institution; policies refreshed for this run |
| BIN | binId: 01a14392-771a-7064-a29c-550f9733f703, binNumber: 433003, binLength: 6, panLength: 16, cardNetwork: VISA, binType: CREDIT, binStatus: ACTIVE |
| 8-digit BIN | binId: 01a14392-7790-7e8f-ba8a-9d9793fbe34d, binNumber: 43253482, binLength: 8, cardNetwork: VISA |
| Issuing PAN range | binRangeId: 01a14392-773a-782b-b51c-a7efff2d8aba, rangeStart: 4330030000000000, rangeEnd: 4330030000009999, allocationStrategy: RANDOM |
| Issuing product | productId: 01a14392-77ec-7af8-99c1-28e464e1ab0f, productCode: VIRTUAL_MAIN_MVAMXDWE, cardNetwork: VISA, fundingType: CREDIT |
| Issuing product version | productVersionId: 01a14392-77f3-7f43-98f4-9ace85e5616f, versionNumber: 1, validityMonths: 60 |
| Virtual card design | cardDesignId: 01a14392-77d1-786a-82a1-e31dca547858, designCode: LFV_MVAMXDWE, designVersion: 1 |
| Physical card design | cardDesignId: 01a14392-77de-7f68-9db6-3e0557efb814, designCode: LFP_MVAMXDWE, designVersion: 1 |
| PAN encryption key | keyId: 01a14392-7c4b-72c0-9ee4-6506514fa25c, keyType: DEK, keyIndex: INST_DEK_a51bbbc1, keyVersion: 14, keyScope: INSTITUTION |
| BIN CVV key (CVK) | keyId: 01a14392-7d6f-768e-bbdc-fe1cc25a7da7, keyType: CVK, keyIndex: BIN_CVK_433003, keyVersion: 1, keyScope: BIN |
| Primary customer | customerReferenceId: eac2cd1d-dc6e-4f5e-a1c3-dfe5265fb40d, externalCustomerId: CUST-MVAMXDWE, sourceSystem: LF_TEST |
| First account | externalAccountReferenceId: 96fe480a-9577-44a0-aaae-f38aba3e1ff4, externalAccountId: ACC-MVAMXDWE-1, accountType: CREDIT_CARD_ACCOUNT |
| Shipping address | addressReferenceId: 8e9a7f30-e65b-430e-991c-ec2c6f1a5455, externalAddressId: ADDR-MVAMXDWE, addressType: SHIPPING, city: Bengaluru, countryCode: IND |
| Configuration-change product (W20–W23) | productId: 01a14395-d111-7ba2-9c56-4a357e7fbca5, productCode: CLIENT_CLIENTCFG_MVAMXDWE |
| Independent product on the same BIN (W24) | productId: 01a14395-f4b2-7431-ab6f-cf10af6d2fa1, productCode: CLIENT_CLIENTISO_MVAMXDWE |
| Cards issued in this run | 39 cards (identifiers listed in the companion file under fixtures.cards_issued) |

The fixture identifiers and 39-card count above describe the archived 16 October baseline. Later verification files identify their own fixtures; the baseline key and card states are historical snapshots.

## 3.2 Complete Switch synchronisation rerun on 20 October 2026

This dated run used WAIT_FOR_SWITCH_SYNC=true from 06:24:15 to 06:32:21 UTC and recorded 1,223 samples, including 18 failing samples. It supplies the successful Switch acknowledgements listed below. Failures also included the closed-card wait, the tenant probe, dependent PIN-fixture errors and the then-unfixed W23 issue. Source: results/switch-sync-verification-2026-10-20.json.

14.10.2 Controls enabled: PASS, settled at Switch revision 7.

19.3.2 Card registration: PASS, settled at Switch revision 2.

19.5.2 Card activation: PASS, settled at Switch revision 3.

19.7.2 Card freeze: PASS, settled at Switch revision 4.

19.9.2 Card release: PASS, settled at Switch revision 5.

19.10 Profile comparison: PASS, exactly one Switch profile and no outstanding command.

19.12.2 Same-PAN successor: PASS, settled at Switch revision 2.

19.13 Separate generations: PASS, source and successor hold separate Switch profiles.

29.5s.2 LOST successor: PASS, settled at Switch revision 2.

18.16.2 Closed card: FAIL, Switch wait timed out at revision 2 with three commands pending. No subsequent successful closure acknowledgement is included in the evidence.

These acknowledgement results apply to this run's fixtures. W04, W17 and W23 were subsequently supplemented or corrected by later runs; current scenario assessments appear in section 1.1. The closed-card timeout remains an open integration item.

### 3.3 PVK and ZPK assignment rerun on 20 October 2026

The later complete run used WAIT_FOR_SWITCH_SYNC=true from 07:37:05 to 07:48:57 UTC and recorded 1,314 samples with seven failing samples: four Switch wait timeouts, the unproven tenant-isolation probe, and the two W23 checks corrected on 21 October. Source: results/pvk-zpk-verification-2026-10-20.json.

Samplers 05.17-05.25 passed: the BIN has ACTIVE PIN_GENERATION and PIN_BLOCK_ENCRYPTION assignments using a BIN PVK and institution ZPK respectively. These nine results are included in the W04 table.

Switch waits were attempted in this run, not skipped. Controls (14.10.2), closed-card status (18.16.2), registration (19.3.2) and LOST-successor registration (29.5s.2) timed out at applied revision 0. Earlier successes on other fixtures do not establish that these later timeouts were resolved.

### 3.4 Hot-list rerun on 20 October 2026

This focused historical hot-list run used WAIT_FOR_SWITCH_SYNC=false from 09:14:00 to 09:14:28 UTC. It recorded 328 samples with no assertion failures and verified CMS hot-list behaviour using operations and cardholder clients. This run-specific setting does not change the true setting used in both complete 20 October runs. Source: results/w17-hotlist-verification-2026-10-20.json.

31.28/31.29 confirmed operations hot-listing and BLOCKED readback; 31.34/31.35 confirmed the same for the cardholder channel. 31.33 verified the customer-only permission. The no-permission case returned 403; CUSTOMER_REQUEST and CLOSED-card business refusals returned 422. W17 includes these later results.

W17 is PASS for CMS hot-listing. Transaction-decline behaviour remains outside the CMS run and is covered separately by the Switch scenarios.

## 4 Scenario results

Scenarios appear in Table 3 order. Tables combine baseline rows with explicitly dated later evidence. S-IDs retain the original check identity when a rerun supersedes a result; added R-references map to dated evidence files in Appendix A. All setup and precondition counts refer to the 16 October baseline unless stated otherwise.

# Phase 1: Institution and security setup

## W01 Create and configure an institution

*Create an institution with a unique code and valid address and contact data. Read it by UUID and by code. Replace its security and operational policies, then read both back. The institution ID and policy ownership must stay consistent. Confirm the accepted country-code format, because the description, the example and the validation pattern disagree in the specification.*

**Result: PASS — 29 of 29 checks passed**

The institution used for the run is the CMS institution mapped to the Switch test institution; it is created or refreshed by fixture 01.1. Create-path validation (duplicate code, invalid code, invalid country) is covered by 32.4 to 32.6. With card issuance disabled in the operational policy, a new issuance is refused at submission (HTTP 422) and existing cards keep working (26.4 to 26.10).

Baseline setup (16 October): 1 card issued and activated (plan steps 26.1, 26.2, 26.3; 6 calls, all passed; data in the companion file under set_up).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0007 | Enable operational policy | PUT /institutions/{inst}/operational-policy | HTTP 200 | PASS |
| S-0008 | Configure security policy | PUT /institutions/{inst}/security-policy | HTTP 200 | PASS |
| S-0279 | A01 Institution lookup by code agrees with lookup by id | GET /institutions/code/FLB_MVAMXDWE | HTTP 200 | PASS |
| S-0592 | A02 Disable card issuance at the institution | PUT /institutions/{inst}/operational-policy | HTTP 200 | PASS |
| S-0594 | Submit rejected issuance: A02 issuance while the institution has issuing disabled | POST /institutions/{inst}/cards | HTTP 422 | PASS |
| S-0595 | Require no card was created: A02 issuance while the institution has issuing disabled | script check | Refused synchronously; no card created | PASS |
| S-0597 | A08 The live card is unaffected by the policy change | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0598 | A08 A live card can still be frozen while issuing is off | POST /institutions/{inst}/cards/{id}/holds/place | HTTP 200 | PASS |
| S-0599 | A08 Release the freeze | POST /institutions/{inst}/cards/{id}/holds/release | HTTP 200 | PASS |
| S-0600 | A02 Restore card issuance at the institution | PUT /institutions/{inst}/operational-policy | HTTP 200 | PASS |
| S-0601 | A02 The operational policy is back to its starting state | GET /institutions/{inst}/operational-policy | HTTP 200 | PASS |
| S-0843 | Institution lookup by UUID | GET /institutions/{inst} | HTTP 200 | PASS |
| S-0844 | Institution list/search | GET /institutions?query=FLB_MVAMXDWE&status=ACTIVE&institutionTyp… | HTTP 200 | PASS |
| S-0845 | Institution partial update | POST /institutions/{inst}/actions/update | HTTP 200 | PASS |
| S-0846 | Duplicate institution code | POST /institutions | HTTP 409 | PASS |
| S-0847 | Invalid institution code characters | POST /institutions | HTTP 400 | PASS |
| S-0848 | Invalid address country code | POST /institutions | HTTP 400 | PASS |
| S-0849 | Security policy retrieval | GET /institutions/{inst}/security-policy | HTTP 200 | PASS |
| S-0850 | Operational policy retrieval | GET /institutions/{inst}/operational-policy | HTTP 200 | PASS |
| S-0851 | Operational policy update then read-back | PUT /institutions/{inst}/operational-policy | HTTP 200 | PASS |
| S-0853 | Invalid base currency | POST /institutions/{inst}/actions/update | HTTP 400 | PASS |
| S-0854 | Timezone longer than 60 characters | POST /institutions/{inst}/actions/update | HTTP 400 | PASS |
| S-0855 | Suspend the institution | POST /institutions/{inst}/actions/update | HTTP 200 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0856 | Restore the institution to ACTIVE | POST /institutions/{inst}/actions/update | HTTP 200 | PASS |
| S-0906 | First operational-policy PUT | PUT /institutions/{inst}/operational-policy | HTTP 200 | PASS |
| S-0907 | Replay operational-policy PUT | PUT /institutions/{inst}/operational-policy | HTTP 200 | PASS |
| S-0908 | First security-policy PUT | PUT /institutions/{inst}/security-policy | HTTP 200 | PASS |
| S-0909 | Replay security-policy PUT | PUT /institutions/{inst}/security-policy | HTTP 200 | PASS |
| S-0912 | Same key on a different endpoint is a separate request | PUT /institutions/{inst}/operational-policy | HTTP 200 | PASS |

## W02 Prepare HSM and institution data keys

*Check provider health. Generate, register or import one suitable protected key, activate it, assign it to the institution for the required data purpose and read the assignment back. Clear key material must never appear in any response, log or audit record.*

### Result: PASS — 19 of 19 checks passed

Generate route used for the PAN encryption and PAN HMAC keys; registration and import routes exercised in 34.5 and 34.8. No clear or wrapped key material appears in any response captured.

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0005 | HSM provider health | GET /hsm/provider-health | HTTP 200 | PASS |
| S-0155 | Generate PAN encryption DEK | POST /institutions/{inst}/hsm-keys | HTTP 201 | PASS |
| S-0156 | Activate PAN encryption DEK | PUT /institutions/{inst}/hsm-keys/{id}/status | HTTP 200 | PASS |
| S-0157 | Create PAN encryption assignment | POST /institutions/{inst}/data-key-assignments | HTTP 201 | PASS |
| S-0158 | Activate PAN encryption assignment | PUT /institutions/{inst}/data-key-assignments/{id}/status | HTTP 200 | PASS |
| S-0159 | Generate PAN HMAC key | POST /institutions/{inst}/hsm-keys | HTTP 201 | PASS |
| S-0160 | Activate PAN HMAC key | PUT /institutions/{inst}/hsm-keys/{id}/status | HTTP 200 | PASS |
| S-0161 | Create PAN HMAC assignment | POST /institutions/{inst}/data-key-assignments | HTTP 201 | PASS |
| S-0162 | Activate PAN HMAC assignment | PUT /institutions/{inst}/data-key-assignments/{id}/status | HTTP 200 | PASS |
| S-0891 | Key list | GET /institutions/{inst}/hsm-keys?page=1&limit=20 | HTTP 200 | PASS |
| S-0892 | Key get | GET /institutions/{inst}/hsm-keys/{id} | HTTP 200 | PASS |
| S-0893 | Data-key assignment list | GET /institutions/{inst}/data-key-assignments?page=1&limit=20 | HTTP 200 | PASS |
| S-0895 | Register an existing key reference | POST /institutions/{inst}/hsm-keys/registrations | HTTP 201 | PASS |
| S-0897 | Incorrect key-purpose scope | POST /institutions/{inst}/data-key-assignments | HTTP 400 | PASS |
| S-0898 | HSM import without KEK fixtures is refused | POST /institutions/{inst}/hsm-keys/imports | HTTP 400 | PASS |
| S-0899 | Create a second PAN encryption assignment | POST /institutions/{inst}/data-key-assignments | HTTP 201 | PASS |
| S-0900 | Replay the successor assignment create | POST /institutions/{inst}/data-key-assignments | HTTP 201 | PASS |
| S-0901 | Activate the successor assignment | PUT /institutions/{inst}/data-key-assignments/{id}/status | HTTP 200 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0902 | Previous PAN assignment is DECRYPT_ONLY | GET /institutions/{inst}/data-key-assignments?keyPurpose=PAN_DATA_ENCRYPTION&page=1&limit=20 | HTTP 200 | PASS |

## W03 Configure BIN and BIN range

*Create a 6 or 8 digit BIN for RUPAY. Read it, replace its mutable configuration, add a non-overlapping same-prefix BIN range, activate the range and verify capacity. Confirm an overlapping range is rejected. Two items to settle here: binType currently offers CREDIT only, so confirm whether debit BINs are in scope, and confirm the accepted format of effectiveFrom, which is typed as a date but shown with a time.*

**Result: PASS — 36 of 36 checks passed**

BINs are issued on the VISA network in this environment; RUPAY-specific BIN configuration was not available. 6-digit and 8-digit BINs, range activation, capacity, overlap rejection, optimistic locking and immutable-field protection were verified.

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0009 | Create BIN | POST /institutions/{inst}/bins | HTTP 201 | PASS |
| S-0010 | Create issuing PAN range | POST /institutions/{inst}/bins/{id}/ranges | HTTP 201 | PASS |
| S-0011 | Submit issuing range | POST /institutions/{inst}/bins/{id}/ranges/{id}/actions/update | HTTP 200 | PASS |
| S-0012 | Activate issuing range | POST /institutions/{inst}/bins/{id}/ranges/{id}/actions/update | HTTP 200 | PASS |
| S-0013 | Create unassigned PAN range | POST /institutions/{inst}/bins/{id}/ranges | HTTP 201 | PASS |
| S-0014 | Submit unassigned range | POST /institutions/{inst}/bins/{id}/ranges/{id}/actions/update | HTTP 200 | PASS |
| S-0015 | Activate unassigned range | POST /institutions/{inst}/bins/{id}/ranges/{id}/actions/update | HTTP 200 | PASS |
| S-0016 | Create an 8-digit BIN | POST /institutions/{inst}/bins | HTTP 201 | PASS |
| S-0859 | BIN get | GET /institutions/{inst}/bins/{id} | HTTP 200 | PASS |
| S-0861 | BIN list/filter page | GET /institutions/{inst}/bins?cardNetwork=VISA&binStatus=ACTIVE&page=1&limit=… | HTTP 200 | PASS |
| S-0862 | BIN list/filter includes both BINs created in this run | script check | Read 1 page(s) of the filtered list; 2 of 2 BIN(s) created in this run are listed | PASS |
| S-0863 | BIN lookup by network | GET /institutions/{inst}/bins/network/VISA?page=1&limit=20 | HTTP 200 | PASS |
| S-0865 | Active BIN lookup by network page | GET /institutions/{inst}/bins/network/VISA/active?page=1&limit=100 | HTTP 200 | PASS |
| S-0866 | Active BIN lookup by network includes the BIN created in this run | script check | Read 1 page(s) of the filtered list; 1 of 1 BIN(s) created in this run are listed | PASS |
| S-0867 | Eight-digit BIN | GET /institutions/{inst}/bins/{id} | HTTP 200 | PASS |
| S-0868 | BIN full PUT with current version | PUT /institutions/{inst}/bins/{id} | HTTP 200 | PASS |
| S-0869 | Optimistic-lock conflict with stale version | PUT /institutions/{inst}/bins/{id} | HTTP 409 | PASS |
| S-0870 | Immutable BIN field rejected | PUT /institutions/{inst}/bins/{id} | HTTP 400 | PASS |
| S-0871 | Invalid binLength | POST /institutions/{inst}/bins | HTTP 400 | PASS |
| S-0872 | Overlapping range on same BIN | POST /institutions/{inst}/bins/{id}/ranges | HTTP 400 | PASS |
| S-0873 | Invalid range boundaries | POST /institutions/{inst}/bins/{id}/ranges | HTTP 400 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0874 | Range list and get | GET /institutions/{inst}/bins/{id}/ranges?page=1&limit=20 | HTTP 200 | PASS |
| S-0875 | Range get | GET /institutions/{inst}/bins/{id}/ranges/{id} | HTTP 200 | PASS |
| S-0876 | Create suspendable range | POST /institutions/{inst}/bins/{id}/ranges | HTTP 201 | PASS |
| S-0877 | Replay suspendable range create | POST /institutions/{inst}/bins/{id}/ranges | HTTP 201 | PASS |
| S-0878 | Submit suspendable range | POST /institutions/{inst}/bins/{id}/ranges/{id}/actions/update | HTTP 200 | PASS |
| S-0879 | Activate suspendable range | POST /institutions/{inst}/bins/{id}/ranges/{id}/actions/update | HTTP 200 | PASS |
| S-0880 | Suspend range | POST /institutions/{inst}/bins/{id}/ranges/{id}/actions/update | HTTP 200 | PASS |
| S-0882 | Submit rejected issuance: issuance from a SUSPENDED range | POST /institutions/{inst}/cards | HTTP 422 | PASS |
| S-0883 | Require no card was created: issuance from a SUSPENDED range | script check | Refused synchronously; no card created | PASS |
| S-0884 | Close the suspended range | POST /institutions/{inst}/bins/{id}/ranges/{id}/actions/update | HTTP 200 | PASS |
| S-0885 | CLOSED range cannot return to ACTIVE | POST /institutions/{inst}/bins/{id}/ranges/{id}/actions/update | HTTP 400 | PASS |
| S-0887 | Submit rejected issuance: issuance from a CLOSED range | POST /institutions/{inst}/cards | HTTP 422 | PASS |
| S-0888 | Require no card was created: issuance from a CLOSED range | script check | Refused synchronously; no card created | PASS |
| S-0916 | Create a disposable BIN | POST /institutions/{inst}/bins | HTTP 201 | PASS |
| S-0917 | Replay disposable BIN create | POST /institutions/{inst}/bins | HTTP 201 | PASS |

## W04 Configure HSM keys for the BIN

*Prepare compatible PVK, CVK, ZPK and EMV keys, activate them, and verify the required BIN assignments. Confirm the PVK and ZPK assignments are ACTIVE before exercising dependent PIN operations.*

**Result: PASS - all 18 listed checks passed: nine baseline checks and nine PVK/ZPK checks from 20 October.**

Baseline CVK/EMV configuration and the later PVK/ZPK verification are combined below. R04 references map to samplers 05.17-05.25 in the dated PVK/ZPK evidence file.

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0163 | Generate BIN CVV key | POST /institutions/{inst}/hsm-keys | HTTP 201 | PASS |
| S-0164 | Activate BIN CVV key | PUT /institutions/{inst}/hsm-keys/{id}/status | HTTP 200 | PASS |
| S-0165 | Create BIN CVV assignment | POST /institutions/{inst}/bins/{id}/hsm-key-assignments | HTTP 201 | PASS |
| S-0166 | Activate BIN CVV assignment | PUT /institutions/{inst}/bins/{id}/hsm-key-assignments/{id}/status | HTTP 200 | PASS |
| S-0167 | Generate EMV IMK (PGK) | POST /institutions/{inst}/hsm-keys | HTTP 201 | PASS |
| S-0168 | Activate EMV IMK | PUT /institutions/{inst}/hsm-keys/{id}/status | HTTP 200 | PASS |
| S-0169 | Create EMV key-derivation assignment | POST /institutions/{inst}/bins/{id}/hsm-key-assignments | HTTP 201 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0170 | Activate EMV key-derivation assignment | PUT /institutions/{inst}/bins/{id}/hsm-key-assignments/{id}/status | HTTP 200 | PASS |
| S-0894 | BIN HSM assignment list | GET /institutions/{inst}/bins/{id}/hsm-key-assignments?page=1&limit=20 | HTTP 200 | PASS |
| R04-17 | Generate BIN PVK | POST /institutions/{inst}/hsm-keys | 20 October rerun: PASS | PASS |
| R04-18 | Activate BIN PVK | PUT /institutions/{inst}/hsm-keys/{id}/status | 20 October rerun: PASS | PASS |
| R04-19 | Create PIN generation assignment | POST /institutions/{inst}/bins/{id}/hsm-key-assignments | 20 October rerun: PASS | PASS |
| R04-20 | Activate PIN generation assignment | PUT /institutions/{inst}/bins/{id}/hsm-key-assignments/{id}/status | 20 October rerun: PASS | PASS |
| R04-21 | Generate institution ZPK | POST /institutions/{inst}/hsm-keys | 20 October rerun: PASS | PASS |
| R04-22 | Activate institution ZPK | PUT /institutions/{inst}/hsm-keys/{id}/status | 20 October rerun: PASS | PASS |
| R04-23 | Create PIN block encryption assignment | POST /institutions/{inst}/bins/{id}/hsm-key-assignments | 20 October rerun: PASS | PASS |
| R04-24 | Activate PIN block encryption assignment | PUT /institutions/{inst}/bins/{id}/hsm-key-assignments/{id}/status | 20 October rerun: PASS | PASS |
| R04-25 | Verify active PVK and ZPK assignments | GET /institutions/{inst}/bins/{id}/hsm-key-assignments | 20 October rerun: PASS; both assignments ACTIVE | PASS |

# Phase 2: Product setup

## W05 Create a product and its first version

*Create the stable card product, verify it by code, create version 1 and read the version. The product ID must stay stable while the version receives its own immutable ID and a positive version number.*

**Result: PASS — 7 of 7 checks passed**

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0023 | Create issuing product | POST /institutions/{inst}/products | HTTP 201 | PASS |
| S-0024 | Create issuing product version | POST /institutions/{inst}/products/{id}/versions | HTTP 201 | PASS |
| S-0604 | Product lookup by UUID | GET /institutions/{inst}/products/{id} | HTTP 200 | PASS |
| S-0605 | Product lookup by code | GET /institutions/{inst}/products/code/VIRTUAL_MAIN_MVAMXDWE | HTTP 200 | PASS |
| S-0606 | Product list and filter | GET /institutions/{inst}/products?query=VIRTUAL_MAIN&cardNetwork=VISA&productStat… | HTTP 200 | PASS |
| S-0620 | Create a new product version after activation | POST /institutions/{inst}/products/{id}/versions | HTTP 201 | PASS |
| S-0621 | Product versions list includes both | GET /institutions/{inst}/products/{id}/versions?page=1&limit=20 | HTTP 200 | PASS |

## W06 Configure product policies and limits

*While the version is draft, set its card-form, PIN, default-control and personalisation policies. Add the required PER_TRANSACTION limit defaults and reason-specific replacement policies. Read everything back before activation. Confirm a default amount above the maximum is rejected, confirm the accepted limit status values, and confirm writes to a version that is no longer draft are refused.*

**Result: PASS — 36 of 36 checks passed**

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0025 | Configure issuing product card form | PUT /institutions/{inst}/products/{id}/versions/{id}/card-form-policy | HTTP 200 | PASS |
| S-0026 | Configure issuing product PIN policy | PUT /institutions/{inst}/products/{id}/versions/{id}/pin-policy | HTTP 200 | PASS |
| S-0027 | Configure issuing product default controls | PUT /institutions/{inst}/products/{id}/versions/{id}/default-controls | HTTP 200 | PASS |
| S-0028 | Create DOMESTIC_ATM limit on issuing product | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-0029 | Create DOMESTIC_POS limit on issuing product | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-0030 | Create DOMESTIC_ECOMMERCE limit on issuing product | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-0031 | Create DOMESTIC_CONTACTLESS limit on issuing product | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-0032 | Create INTERNATIONAL_ATM limit on issuing product | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-0033 | Create INTERNATIONAL_POS limit on issuing product | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-0034 | Create INTERNATIONAL_ECOMMERCE limit on issuing product | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-0035 | Create INTERNATIONAL_CONTACTLESS limit on issuing product | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-0036 | Create RENEWAL replacement policy | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies | HTTP 201 | PASS |
| S-0037 | Create DAMAGED replacement policy | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies | HTTP 201 | PASS |
| S-0038 | Create LOST replacement policy | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies | HTTP 201 | PASS |
| S-0039 | Create STOLEN replacement policy | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies | HTTP 201 | PASS |
| S-0040 | Create COMPROMISED replacement policy | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies | HTTP 201 | PASS |
| S-0041 | Create UPGRADE (disallowed) replacement policy | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies | HTTP 201 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0042 | Configure issuing product personalisation policy | PUT /institutions/{inst}/products/{id}/versions/{id}/personalisation-policy | HTTP 200 | PASS |
| S-0607 | Default controls read API | GET /institutions/{inst}/products/{id}/versions/{id}/default-controls | HTTP 200 | PASS |
| S-0608 | Personalisation policy read | GET /institutions/{inst}/products/{id}/versions/{id}/personalisation-policy | HTTP 200 | PASS |
| S-0609 | Replacement policies list | GET /institutions/{inst}/products/{id}/versions/{id}/replacement-policies?page=1&limit=20 | HTTP 200 | PASS |
| S-0610 | Replacement policy by id | GET /institutions/{inst}/products/{id}/versions/{id}/replacement-policies/{id} | HTTP 200 | PASS |
| S-0611 | Product limits list and get | GET /institutions/{inst}/products/{id}/versions/{id}/limits?page=1&limit=20 | HTTP 200 | PASS |
| S-0612 | Individual limit retrieval | GET /institutions/{inst}/products/{id}/versions/{id}/limits/{id} | HTTP 200 | PASS |
| S-0618 | Activated version configuration is immutable | POST /institutions/{inst}/products/{id}/versions/{id}/actions/update | HTTP 400 | PASS |
| S-0619 | Personalisation cannot change on an ACTIVE version | PUT /institutions/{inst}/products/{id}/versions/{id}/personalisation-policy | HTTP 400 | PASS |
| S-0623 | Replacement policy update only while DRAFT | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies/{id}/actions/update | HTTP 400 | PASS |
| S-0624 | Replacement policy update on the draft product | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies | HTTP 201 | PASS |
| S-0625 | Update draft replacement policy | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies/{id}/actions/update | HTTP 200 | PASS |
| S-0640 | Limit update on an ACTIVE version is refused | POST /institutions/{inst}/products/{id}/versions/{id}/limits/{id}/actions/update | HTTP 400 | PASS |
| S-0641 | Create a limit on the DRAFT version | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-0642 | Update the DRAFT limit | POST /institutions/{inst}/products/{id}/versions/{id}/limits/{id}/actions/update | HTTP 200 | PASS |
| S-0643 | Overlapping effective dates stay a second DRAFT | POST /institutions/{inst}/products/{id}/versions | HTTP 201 | PASS |
| S-0644 | The product still has one ACTIVE version | GET /institutions/{inst}/products/{id}/versions?page=1&limit=20 | HTTP 200 | PASS |
| S-0938 | Replacement-policy first page | GET /institutions/{inst}/products/{id}/versions/{id}/replacement-policies?page=1&limit=2 | HTTP 200 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0939 | Replacement-policy middle page | GET /institutions/{inst}/products/{id}/versions/{id}/replacement-policies?page=2&limit=2 | HTTP 200 | PASS |

## W07 Configure design, technology and CVM

*Create and activate a card design. Create and activate the product version technology profile and CVM profile. CVM rules must have unique positive sequence numbers and use supported interface, method and condition values, and must be evaluated in ascending sequence.*

**Result: PASS — 21 of 21 checks passed**

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0018 | Create virtual card design | POST /institutions/{inst}/card-designs | HTTP 201 | PASS |
| S-0019 | Activate virtual card design | POST /institutions/{inst}/card-designs/{id}/actions/update | HTTP 200 | PASS |
| S-0020 | Create physical card design | POST /institutions/{inst}/card-designs | HTTP 201 | PASS |
| S-0021 | Activate physical card design | POST /institutions/{inst}/card-designs/{id}/actions/update | HTTP 200 | PASS |
| S-0046 | Create issuing product CVM profile | POST /institutions/{inst}/products/{id}/versions/{id}/cvm-profiles | HTTP 201 | PASS |
| S-0047 | Activate issuing product CVM profile | POST /institutions/{inst}/products/{id}/versions/{id}/cvm-profiles/{id}/actions/update | HTTP 200 | PASS |
| S-0104 | Create physical-only product technology profile | POST /institutions/{inst}/products/{id}/versions/{id}/technology-profiles | HTTP 201 | PASS |
| S-0105 | Activate physical-only product technology profile | POST /institutions/{inst}/products/{id}/versions/{id}/technology-profiles/{id}/actions/update | HTTP 200 | PASS |
| S-0613 | Card design get | GET /institutions/{inst}/card-designs/{id} | HTTP 200 | PASS |
| S-0614 | Card design list | GET /institutions/{inst}/card-designs?cardNetwork=VISA&designStatus=ACTIVE&page=1&lim… | HTTP 200 | PASS |
| S-0615 | Design assignment list | GET /institutions/{inst}/products/{id}/versions/{id}/card-design-assignments?page=1&limit=20 | HTTP 200 | PASS |
| S-0616 | Technology profile get | GET /institutions/{inst}/products/{id}/versions/{id}/technology-profiles/{id} | HTTP 200 | PASS |
| S-0617 | CVM profile get with ordered rules | GET /institutions/{inst}/products/{id}/versions/{id}/cvm-profiles/{id} | HTTP 200 | PASS |
| S-0626 | Create a disposable card design | POST /institutions/{inst}/card-designs | HTTP 201 | PASS |
| S-0627 | Rename a DRAFT design | POST /institutions/{inst}/card-designs/{id}/actions/update | HTTP 200 | PASS |
| S-0628 | Activate the disposable design | POST /institutions/{inst}/card-designs/{id}/actions/update | HTTP 200 | PASS |
| S-0629 | ACTIVE design configuration is immutable | POST /institutions/{inst}/card-designs/{id}/actions/update | HTTP 400 | PASS |
| S-0630 | Suspend the disposable design | POST /institutions/{inst}/card-designs/{id}/actions/update | HTTP 200 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0631 | Reactivate the disposable design | POST /institutions/{inst}/card-designs/{id}/actions/update | HTTP 200 | PASS |
| S-0632 | Retire the disposable design | POST /institutions/{inst}/card-designs/{id}/actions/update | HTTP 200 | PASS |
| S-0633 | RETIRED design cannot return to ACTIVE | POST /institutions/{inst}/card-designs/{id}/actions/update | HTTP 400 | PASS |

## W08 Link the range and design to product, then activate

*Assign an active institution-owned BIN range to the product. Assign the active design to the product version for the required card form. Activate both assignments, then move the product and the version to ACTIVE. Confirm activation is rejected when a required assignment or profile is missing. Record the legal status transitions as you go,*

### Result: PASS — 24 of 24 checks passed

Baseline preconditions (16 October): steps 04.1 to 04.8 (6 calls, all passed; data in the companion file).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0043 | Assign BIN range to issuing product | POST /institutions/{inst}/products/{id}/bin-range-assignments | HTTP 201 | PASS |
| S-0044 | Assign virtual card design to issuing product | POST /institutions/{inst}/products/{id}/versions/{id}/card-design-assignments | HTTP 201 | PASS |
| S-0045 | Activate issuing product design assignment | POST /institutions/{inst}/products/{id}/versions/{id}/card-design-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-0048 | Submit issuing product | POST /institutions/{inst}/products/{id}/actions/update | HTTP 200 | PASS |
| S-0049 | Activate issuing product | POST /institutions/{inst}/products/{id}/actions/update | HTTP 200 | PASS |
| S-0050 | Submit issuing product version | POST /institutions/{inst}/products/{id}/versions/{id}/actions/update | HTTP 200 | PASS |
| S-0051 | Activate issuing product version | POST /institutions/{inst}/products/{id}/versions/{id}/actions/update | HTTP 200 | PASS |
| S-0052 | Submit issuing product assignment | POST /institutions/{inst}/products/{id}/bin-range-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-0053 | Activate issuing product assignment | POST /institutions/{inst}/products/{id}/bin-range-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-0260 | A06 Reject DRAFT to ACTIVE in one step | POST /institutions/{inst}/products/{id}/versions/{id}/actions/update | HTTP 400 | PASS |
| S-0261 | A06 Draft version is unchanged | GET /institutions/{inst}/products/{id}/versions/{id} | HTTP 200 | PASS |
| S-0263 | Submit rejected issuance: A05 issuance on a DRAFT product version | POST /institutions/{inst}/cards | HTTP 422 | PASS |
| S-0264 | Require no card was created: A05 issuance on a DRAFT product version | script check | Refused synchronously; no card created | PASS |
| S-0266 | Submit rejected issuance: A03 issuance on an unassigned PAN range | POST /institutions/{inst}/cards | HTTP 422 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0267 | Require no card was created: A03 issuance on an unassigned PAN range | script check | Refused synchronously; no card created | PASS |
| S-0269 | Submit rejected issuance: A03 issuance on an unknown PAN range | POST /institutions/{inst}/cards | HTTP 404 | PASS |
| S-0270 | Require no card was created: A03 issuance on an unknown PAN range | script check | Refused synchronously; no card created | PASS |
| S-0622 | Draft version cannot jump straight to ACTIVE | POST /institutions/{inst}/products/{id}/versions/{id}/actions/update | HTTP 400 | PASS |
| S-0634 | Assign the virtual design to the DRAFT product | POST /institutions/{inst}/products/{id}/versions/{id}/card-design-assignments | HTTP 201 | PASS |
| S-0635 | Activate the draft-product design assignment | POST /institutions/{inst}/products/{id}/versions/{id}/card-design-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-0636 | ACTIVE assignment configuration is immutable | POST /institutions/{inst}/products/{id}/versions/{id}/card-design-assignments/{id}/actions/update | HTTP 400 | PASS |
| S-0637 | Suspend the draft-product design assignment | POST /institutions/{inst}/products/{id}/versions/{id}/card-design-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-0638 | Close the draft-product design assignment | POST /institutions/{inst}/products/{id}/versions/{id}/card-design-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-0639 | CLOSED assignment cannot return to ACTIVE | POST /institutions/{inst}/products/{id}/versions/{id}/card-design-assignments/{id}/actions/update | HTTP 400 | PASS |

# Phase 3: Card issuance

## W09 Issue a virtual-only card

*Use an active VIRTUAL_ONLY version and valid customer, account, product, version and range references. Verify virtual-only issuance, the absence of a physical-production requirement, and a masked PAN with last four only. Card activation is available and is verified in the lifecycle checks.*

**Result: PASS — 36 of 36 checks passed**

Issuance is asynchronous: the submit call returns HTTP 202 with an operation id; the plan polls the operation and then requires the card to exist with a masked PAN.

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0272 | Submit issuance: A09 virtual card | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0275 | Read issuance operation: A09 virtual card | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-0276 | Require issued card: A09 virtual card | script check | Card issued: 01a14392-838e-7e86-bae5-da7a7a5ad28f | PASS |
| S-0277 | A09 Reject physical production on a VIRTUAL_ONLY product | POST /institutions/{inst}/cards/{id}/physical-production-requests | HTTP 400 | PASS |
| S-0278 | A09 The refused request created no production reference | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0283 | Submit rejected issuance: B02 unknown customer reference | POST /institutions/{inst}/cards | HTTP 404 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0284 | Require no card was created: B02 unknown customer reference | script check | Refused synchronously; no card created | PASS |
| S-0286 | Submit rejected issuance: B02 unknown account reference | POST /institutions/{inst}/cards | HTTP 404 | PASS |
| S-0287 | Require no card was created: B02 unknown account reference | script check | Refused synchronously; no card created | PASS |
| S-0289 | Submit issuance: B04 first card on shared account | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0291 | Read issuance operation: B04 first card on shared account | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-0292 | Require issued card: B04 first card on shared account | script check | Card issued: 01a14392-9c27-7a57-a230-7ecd7f52eba9 | PASS |
| S-0294 | Submit issuance: B04 second card on shared account | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0297 | Read issuance operation: B04 second card on shared account | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-0298 | Require issued card: B04 second card on shared account | script check | Card issued: 01a14392-ac23-74be-9cdb-2f5a30a5720a | PASS |
| S-0299 | B04 The two cards are distinct identities | script check | Distinct cards 01a14392-9c27-7a57-a230-7ecd7f52eba9 and 01a14392-ac23-74be-9cdb-2f5a30a5720a | PASS |
| S-0300 | B04 Both cards report the same account linkage | GET /institutions/{inst}/cards?accountReferenceId={id}&page=1&limit=10 | HTTP 200 | PASS |
| S-0301 | Activate B04 first card | POST /institutions/{inst}/cards/{id}/actions/activate | HTTP 200 | PASS |
| S-0302 | B06 Activating one card left the sibling untouched | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0305 | C04 Missing customerReferenceId is rejected | POST /institutions/{inst}/cards | HTTP 400 | PASS |
| S-0306 | C04 Missing accountReferenceId is rejected | POST /institutions/{inst}/cards | HTTP 400 | PASS |
| S-0307 | C04 Missing productId is rejected | POST /institutions/{inst}/cards | HTTP 400 | PASS |
| S-0308 | C04 Missing productVersionId is rejected | POST /institutions/{inst}/cards | HTTP 400 | PASS |
| S-0309 | C04 Missing binRangeId is rejected | POST /institutions/{inst}/cards | HTTP 400 | PASS |
| S-0310 | C04 Missing activationPolicyVersionId is rejected | POST /institutions/{inst}/cards | HTTP 400 | PASS |
| S-0311 | C04 Malformed UUID is rejected | POST /institutions/{inst}/cards | HTTP 400 | PASS |
| S-0312 | C04 Unsupported accountLinkType is rejected | POST /institutions/{inst}/cards | HTTP 400 | PASS |
| S-0313 | C04 Oversized embossed name is rejected | POST /institutions/{inst}/cards | HTTP 400 | PASS |
| S-0314 | C06 Issuance without an Idempotency-Key is rejected | POST /institutions/{inst}/cards | HTTP 400 | PASS |
| S-0324 | C07 Unknown issuance operation is not found | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 404 | PASS |
| S-0524 | R03 Submit the shared issuance request | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0528 | R03 Exactly one card exists for the shared key | GET /institutions/{inst}/cards?accountReferenceId={id}&page=1&limit=10 | HTTP 200 | PASS |
| S-0703 | Submit issuance: virtual-only production probe | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0705 | Read issuance operation: virtual-only production probe | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-0706 | Require issued card: virtual-only production probe | script check | Card issued: 01a14394-e0ff-7898-8173-95f778a3d195 | PASS |
| S-0707 | VIRTUAL_ONLY still refuses physical production | POST /institutions/{inst}/cards/{id}/physical-production-requests | HTTP 422 | PASS |

## W10 Issue a physical-only card

*Use an active PHYSICAL_ONLY version with an eligible design and technology and CVM profiles. The production batch reference is required when the product creates a physical card immediately. The shipping address reference is optional, and production stays address-not-ready when it is omitted, so test both. Verify the physical form and the production request state.*

**Result: PASS — 18 of 18 checks passed**

Card A (41.12) is a physical-only issuance with a shipping address and production batch reference. The address-not-ready variant (no shipping address) is not exercised in this plan.

Baseline preconditions (16 October): steps 04c.1 to 04c.8f (19 calls, all passed; data in the companion file).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0676 | Submit issuance: PHYSICAL_ONLY card | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0678 | Read issuance operation: PHYSICAL_ONLY card | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-0679 | Require issued card: PHYSICAL_ONLY card | script check | Card issued: 01a14394-913f-7e4d-b1a7-545714f79c3a | PASS |
| S-0680 | PHYSICAL_ONLY confirms a physical card and a production request | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0708 | activate-physical before delivery | POST /institutions/{inst}/cards/{id}/commands/activate-physical | HTTP 422 | PASS |
| S-0709 | make-use-ready before physical activation | POST /institutions/{inst}/cards/{id}/commands/make-use-ready | HTTP 422 | PASS |
| S-0711 | Physical-only product stores STANDARD production | GET /institutions/{inst}/products/{id}/versions/{id}/personalisation-policy | HTTP 200 | PASS |
| S-0713 | List personalisation batches on the embossing service | GET /institutions/{inst}/personalisation-batches?page=1&limit=20 | HTTP 200 | PASS |
| S-0714 | List CREATED personalisation batches | GET /institutions/{inst}/personalisation-batches?status=CREATED&page=1&limit=20 | HTTP 200 | PASS |
| S-0715 | Unknown personalisation batch is not found | GET /institutions/{inst}/personalisation-batches/{id} | HTTP 404 | PASS |
| S-0716 | Unknown personalisation file is not found | GET /institutions/{inst}/personalisation-files/{id} | HTTP 404 | PASS |
| S-0717 | Files for an unknown batch are not found | GET /institutions/{inst}/personalisation-batches/{id}/files | HTTP 404 | PASS |
| S-0718 | Invalid batch status is refused | GET /institutions/{inst}/personalisation-batches?status=not-a-status&page=1&limit=20 | HTTP 400 | PASS |
| S-0719 | Read the listed personalisation batch | GET /institutions/{inst}/personalisation-batches/{id} | HTTP 200 | PASS |
| S-0720 | List files for the listed personalisation batch | GET /institutions/{inst}/personalisation-batches/{id}/files | HTTP 200 | PASS |
| S-0998 | Submit issuance: CLIENT CARD A | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-1000 | Read issuance operation: CLIENT CARD A | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-1001 | Require issued card: CLIENT CARD A | script check | Card issued: 01a14395-d22c-790c-86ec-4af5a28e5be4 | PASS |

## W11 Issue virtual first, then request physical

*Use VIRTUAL_THEN_PHYSICAL_LATER. Issue the virtual card, verify the later physical timing, then request physical production with valid shipping and production batch references, both of which are required on this call. The card ID and PAN allocation must be preserved. Confirm a second production request for the same card is rejected.*

**Result: PASS — 21 of 21 checks passed**

Baseline preconditions (16 October): steps 04b.1 to 04b.8f (19 calls, all passed; data in the companion file).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0648 | Submit issuance: virtual-then-physical card | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0651 | Read issuance operation: virtual-then-physical card | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-0652 | Require issued card: virtual-then-physical card | script check | Card issued: 01a14394-5123-7fd6-a1e6-fb60719bfd5f | PASS |
| S-0653 | VIRTUAL_THEN_PHYSICAL_LATER issues as VIRTUAL and is not yet activated | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0654 | Successful physical-production request | POST /institutions/{inst}/cards/{id}/physical-production-requests | HTTP 201 | PASS |
| S-0655 | Replay the physical-production request | POST /institutions/{inst}/cards/{id}/physical-production-requests | HTTP 201 | PASS |
| S-0656 | Duplicate physical-production request | POST /institutions/{inst}/cards/{id}/physical-production-requests | HTTP 422 | PASS |
| S-0658 | Submit issuance: later card for close | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0660 | Read issuance operation: later card for close | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-0661 | Require issued card: later card for close | script check | Card issued: 01a14394-691b-7773-bafc-d67e78848306 | PASS |
| S-0662 | Activate later2 | POST /institutions/{inst}/cards/{id}/actions/activate | HTTP 200 | PASS |
| S-0663 | Close the later card | POST /institutions/{inst}/cards/{id}/actions/close | HTTP 200 | PASS |
| S-0664 | Production after closure | POST /institutions/{inst}/cards/{id}/physical-production-requests | HTTP 422 | PASS |
| S-0666 | Submit issuance: later card for block | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0669 | Read issuance operation: later card for block | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-0670 | Require issued card: later card for block | script check | Card issued: 01a14394-7922-7b71-917e-7959bf9ca40b | PASS |
| S-0671 | Activate later3 | POST /institutions/{inst}/cards/{id}/actions/activate | HTTP 200 | PASS |
| S-0672 | Permanent block before production | POST /institutions/{inst}/cards/{id}/holds/place | HTTP 200 | PASS |
| S-0673 | Production after permanent block | POST /institutions/{inst}/cards/{id}/physical-production-requests | HTTP 422 | PASS |
| S-0674 | Close a BLOCKED card | POST /institutions/{inst}/cards/{id}/actions/close | HTTP 200 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0710 | Later product stores PRIORITY production | GET /institutions/{inst}/products/{id}/versions/{id}/personalisation-policy | HTTP 200 | PASS |

## W12 Issue virtual and physical together

*Use VIRTUAL_AND_PHYSICAL_SAME_DAY. Issue with valid physical production references and verify the combined card form, the same-day physical timing and the production request details.*

**Result: PASS — 5 of 5 checks passed**

Baseline preconditions (16 October): steps 04d.1 to 04d.8f (19 calls, all passed; data in the companion file).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0682 | Submit issuance: same-day virtual and physical card | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0685 | Read issuance operation: same-day virtual and physical card | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-0686 | Require issued card: same-day virtual and physical card | script check | Card issued: 01a14394-a126-7ba5-a13b-c289facc64a2 | PASS |
| S-0687 | SAME_DAY confirms a dual-form card with its own PAN | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0712 | Same-day product stores EMERGENCY production | GET /institutions/{inst}/products/{id}/versions/{id}/personalisation-policy | HTTP 200 | PASS |

## W13 Issue a physical card with multiple virtual PAN capability

*Use PHYSICAL_WITH_MULTIPLE_VIRTUAL_PANS with a positive maximum. Verify the recorded issuance mode and maximum, successful issuance within the configured cap, and refusal beyond the cap. Management of additional PANs on an already-issued card was not exercised in this evidence.*

**Result: PASS — 14 of 14 checks passed**

The policy recorded PHYSICAL_WITH_MULTIPLE_VIRTUAL_PANS and the configured maximum. Issuance beyond the account cap returned HTTP 422 and created no card (28.16c; 12.12/12.13 for a maximum of one). The result covers the recorded configuration and issuance-cap checks.

Baseline preconditions (16 October): steps 03.1 to 04e.8f (31 calls, all passed; data in the companion file).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0316 | Submit issuance: C05 first card at the ceiling | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0318 | Read issuance operation: C05 first card at the ceiling | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-0319 | Require issued card: C05 first card at the ceiling | script check | Card issued: 01a14392-c4d7-72c6-b9fa-3e794a2cd1fc | PASS |
| S-0321 | Submit rejected issuance: C05 second card beyond the single-PAN ceiling | POST /institutions/{inst}/cards | HTTP 422 | PASS |
| S-0322 | Require no card was created: C05 second card beyond the single-PAN ceiling | script check | Refused synchronously; no card created | PASS |
| S-0323 | C05 The ceiling account still holds exactly one card | GET /institutions/{inst}/cards?accountReferenceId={id}&page=1&limit=10 | HTTP 200 | PASS |
| S-0689 | Submit issuance: multi virtual 1 | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0691 | Read issuance operation: multi virtual 1 | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-0692 | Require issued card: multi virtual 1 | script check | Card issued: 01a14394-b900-7cf0-abf3-499abe8e4b6b | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0694 | Submit issuance: multi virtual 2 | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-0697 | Read issuance operation: multi virtual 2 | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-0698 | Require issued card: multi virtual 2 | script check | Card issued: 01a14394-c8f9-7dc8-9020-49f68dec9e13 | PASS |
| S-0700 | Submit rejected issuance: maximumActiveVirtualPans=2 refuses a third card | POST /institutions/{inst}/cards | HTTP 422 | PASS |
| S-0701 | Require no card was created: maximumActiveVirtualPans=2 refuses a third card | script check | Refused synchronously; no card created | PASS |

# Phase 4: Controls and limits

## W14 Change card controls

*Read controls, change enabled flags across domestic/international ATM, POS and online dimensions, and read them back. Verify eventual Switch acknowledgement of the updated control version. Record concurrent-update and idempotency behaviour separately. Transaction authorisation requires the separate Switch scenarios.*

**Result: PASS - all 27 listed checks passed, including the 20 October Switch acknowledgement at revision 7.**

The table combines 26 baseline checks with the dated 14.10.2 acknowledgement. The Switch reached revision 7 on that run's fixture; later timeout results remain separately recorded in section 3.3.

Baseline setup (16 October): 1 card issued and activated (plan steps 14.1, 14.2; 6 calls, all passed; data in the companion file under set_up).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0345 | H09 A blocked card cannot change controls | PUT /institutions/{inst}/cards/{id}/controls | HTTP 422 | PASS |
| S-0356 | H05 The card starts on product defaults | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0357 | H08 More than eight dimensions is rejected | PUT /institutions/{inst}/cards/{id}/controls | HTTP 400 | PASS |
| S-0358 | H08 A dimension without enabled is rejected | PUT /institutions/{inst}/cards/{id}/controls | HTTP 400 | PASS |
| S-0359 | H08 An unknown channel is rejected | PUT /institutions/{inst}/cards/{id}/controls | HTTP 400 | PASS |
| S-0360 | H08 An unknown geography is rejected | PUT /institutions/{inst}/cards/{id}/controls | HTTP 400 | PASS |
| S-0361 | H08 A negative per-transaction limit is rejected | PUT /institutions/{inst}/cards/{id}/controls | HTTP 400 | PASS |
| S-0362 | H08 A null controls list is rejected | PUT /institutions/{inst}/cards/{id}/controls | HTTP 400 | PASS |
| S-0363 | H08 A repeated dimension in one request is rejected | PUT /institutions/{inst}/cards/{id}/controls | HTTP 422 | PASS |
| S-0364 | H08 No invalid payload changed the profile | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0365 | H01 Enable the international master switch | PUT /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0366 | H03 Enable one dimension only | PUT /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0367 | H01 Enable dimension batch 1 | PUT /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0368 | H01 Enable dimension batch 2 | PUT /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0369 | H01 The international scope survived the dimension sweep | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0370 | H09 Every dimension reads back enabled | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0371 | H02 Disable dimension batch 1 | PUT /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0372 | H02 Disable dimension batch 2 | PUT /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0373 | H09 Every dimension reads back disabled | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0374 | H02 A repeated no-op update is accepted | PUT /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0464 | M07 A closed card cannot change controls | PUT /institutions/{inst}/cards/{id}/controls | HTTP 422 | PASS |
| S-0509 | R01 First control update | PUT /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0510 | R01 Replay the control update | PUT /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0511 | R01 Conflicting control payload under the same key | PUT /institutions/{inst}/cards/{id}/controls | HTTP 409 | PASS |
| S-0512 | R04 A control update without an Idempotency-Key is rejected | PUT /institutions/{inst}/cards/{id}/controls | HTTP 400 | PASS |
| S-0513 | R04 A malformed Idempotency-Key is rejected | PUT /institutions/{inst}/cards/{id}/controls | HTTP 400 | PASS |
| R14-ACK | Wait for Switch to apply enabled controls | GET /institutions/{inst}/cards/{id}/switch-sync | 20 October 06:24 run: PASS; settled at Switch revision 7 | PASS |

## W15 Change per-transaction limits

*Confirm the product maximums, then set card-level per-transaction values within the product ceiling and the Accounting assigned limit. For POS also test contactless enabled and the contactless limit. Verify the returned control version and values, and verify the Switch applies them. The Accounting figure is a snapshot the specification calls non-authoritative, so also test with a deliberately stale snapshot and record the error returned. The authorisation proof is workflows 26 to 35.*

**Result: PARTIAL - all nine executed rows passed; the final authorisation case (TC-015-05) remains unverified due to the simulator limitation.**

The 21 October focused run used the local Accounting stub, verified the ceiling, one-unit-over and stale-snapshot cases, and confirmed Switch profile revision 7. TC-015-04 is split into four rows below. S-0361 and S-0384 are baseline checks. Live Accounting integration was not verified.

Baseline setup (16 October): 1 card issued and activated (plan steps 15.1, 15.2; 5 calls, all passed; data in the companion file under set_up).

**Known test-bed limitation.** The NPCI simulator is configured with a different transaction card, so final authorisation using the CMS scenario card cannot be validated. Limit enforcement below, at and above the configured limit remains unverified (TC-015-05).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0361 | H08 A negative per-transaction limit is rejected | PUT /institutions/{inst}/cards/{id}/controls | HTTP 400 | PASS |
| S-0384 | I01 The issued card carries the product limits | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| TC-015-01 | Accept the exact ceiling | PUT /institutions/{inst}/cards/{id}/controls | Domestic e-commerce was set to 100000 minor units (INR 1,000), the product ceiling. CMS stored 100000, and Switch acknowledged that profile. | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| TC-015-02 | Reject one unit above the ceiling | PUT /institutions/{inst}/cards/{id}/controls | A limit of 100001 was refused with HTTP 422. The stored limit stayed 100000. | PASS |
| TC-015-03 | Reject a stale Accounting snapshot | PUT /institutions/{inst}/cards/{id}/controls | The Accounting snapshot was older than the allowed window. The update was refused with HTTP 400. The stored limit stayed 100000. | PASS |
| TC-015-04a | Store zero as an explicit limit | PUT /institutions/{inst}/cards/{id}/controls | Domestic e-commerce was set to 0. CMS stored 0, and the final Switch profile holds 0. | PASS |
| TC-015-04b | Keep a disabled channel disabled | PUT /institutions/{inst}/cards/{id}/controls | Domestic MOTO was left disabled and given a limit of 100000. The channel stayed disabled. The final Switch profile holds the same state. | PASS |
| TC-015-04c | Store contactless on its own row | PUT /institutions/{inst}/cards/{id}/controls | Contactless was stored on its own CONTACTLESS_POS row, disabled, at 5000. Domestic ATM stayed disabled at 50000. | PASS |
| TC-015-04d | Reject a nested contactless limit | PUT /institutions/{inst}/cards/{id}/controls | A contactless limit sent inside a point-of-sale row was refused with HTTP 422. | PASS |
| TC-015-05 | Authorise below, at and above the limit | Not sent | Known test-bed limitation: the NPCI simulator uses a different card. Limit approval and decline behaviour remains unverified. | NOT EXECUTED |

## Phase 5: PIN and hot-listing

### W16 Change and reset PIN

*Enable both operations in the active PIN policy and confirm the BIN has an active ZPK. Change the PIN using current and new encrypted format-0 blocks, read history, then reset it with a new encrypted block and a reason. No PIN block, clear PIN or verification offset may be returned or logged. Confirm the accepted values for PIN method and unblock method, because the examples in the specification are not in the allowed lists. Confirm a wrong current PIN returns the correct error. Online PIN is proved at authorisation in workflows 28 and 30.*

**Result: PARTIAL - 14 validation, metadata and refusal checks passed; successful PIN operations and final authorisation remain unverified.**

Successful PIN selection, change and reset were not exercised because the test plan used placeholder PIN blocks.

Baseline setup (16 October): 1 card issued and activated (plan steps 30.1, 30.2; 5 calls, all passed; data in the companion file under set_up).

**Known test-bed limitation.** The NPCI simulator is configured with a different transaction card, so final authorisation using the CMS scenario card cannot be validated. Approval with the new PIN and refusal with the old PIN remain unverified.

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0419 | A malformed PIN block is a validation error | POST /institutions/{inst}/cards/{id}/pin/actions/change | HTTP 400 | PASS |
| S-0420 | PIN reset must not include the current PIN | POST /institutions/{inst}/cards/{id}/pin/actions/reset | HTTP 400 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0789 | PIN history is readable without PIN material | GET /institutions/{inst}/cards/{id}/pin/history ?page=1&limit=20 | HTTP 200 | PASS |
| S-0790 | PIN state is metadata for a Switch-enrolled family | GET /institutions/{inst}/cards/{id}/pin | HTTP 200 | PASS |
| S-0791 | A PIN change with a short block is a validation error | POST /institutions/{inst}/cards/{id}/pin/actions /change | HTTP 400 | PASS |
| S-0792 | PIN change without the current PIN is refused | POST /institutions/{inst}/cards/{id}/pin/actions /change | HTTP 400 | PASS |
| S-0793 | An unsupported PIN block format is a validation error | POST /institutions/{inst}/cards/{id}/pin/actions /initialize | HTTP 400 | PASS |
| S-0794 | A PIN change is refused before a PIN is established | POST /institutions/{inst}/cards/{id}/pin/actions /change | HTTP 409 | PASS |
| S-0795 | First PIN selection cannot claim an existing PIN version | POST /institutions/{inst}/cards/{id}/pin/actions /initialize | HTTP 409 | PASS |
| S-0796 | Place a permanent LOST hold on the PIN card | POST /institutions/{inst}/cards/{id}/holds/plac e | HTTP 200 | PASS |
| S-0797 | A permanent security hold refuses PIN change | POST /institutions/{inst}/cards/{id}/pin/actions /change | HTTP 422 | PASS |
| S-0798 | Close the blocked PIN card | POST /institutions/{inst}/cards/{id}/actions/clo se | HTTP 200 | PASS |
| S-0799 | A closed card refuses PIN reset | POST /institutions/{inst}/cards/{id}/pin/actions /reset | HTTP 422 | PASS |
| S-0800 | An unknown PIN operation is not found | GET /institutions/{inst}/cards/{id}/pin/operati ons/{id} | HTTP 404 | PASS |

## W17 Hot-list a card

*Needs the Switch test endpoint. Hot-list an active card with a supported reason. Verify the response records the prior status, BLOCKED status, event ID, caller type and lifecycle version, and that the card read shows the blocked state. Run it twice, once with a cardholder-channel token and once with an operations token, and confirm the caller type is taken from the token and not from the payload. The decline proof is workflow 35.*

**Result: PASS - all 12 listed checks passed, including operations/cardholder hot-listing and corrected 422 business-refusal results from 20 October.**

The table includes four baseline hold checks and dated hot-list permission, business-refusal, positive-channel and readback evidence. Transaction decline after hot-listing was not verified in this CMS run.

Baseline setup (16 October): 2 cards issued and activated (plan steps 13.5, 13.6, 31.1, 31.2; 10 calls, all passed; data in the companion file under set_up).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0342 | D04 Report the card LOST | POST /institutions/{inst}/cards/{id}/holds/plac e | HTTP 200 | PASS |
| S-0343 | D04 A blocked card cannot be activated | POST /institutions/{inst}/cards/{id}/actions/act ivate | HTTP 422 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0344 | G03 A permanent LOST hold cannot be released | POST /institutions/{inst}/cards/{id}/holds/release | HTTP 422 | PASS |
| S-0346 | D04 The refused actions left the card blocked | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0819 | Hotlist without hotlist permission | POST /institutions/{inst}/cards/{id}/hotlist | 20 October rerun: HTTP 403 ACCESS_DENIED | PASS |
| S-0838 | Hotlist CUSTOMER_REQUEST is refused before Switch | POST /institutions/{inst}/cards/{id}/hotlist | 20 October rerun: HTTP 422 business-rule refusal | PASS |
| S-0839 | Hotlist of a CLOSED card is refused before Switch | POST /institutions/{inst}/cards/{id}/hotlist | 20 October rerun: HTTP 422 business-rule refusal | PASS |
| R17-28 | Hot-list active card as operations | POST /institutions/{inst}/cards/{id}/hotlist | 20 October rerun: HTTP 200; callerType OPERATIONS | PASS |
| R17-29 | Read operations-hotlisted card | GET /institutions/{inst}/cards/{id} | 20 October rerun: HTTP 200; BLOCKED | PASS |
| R17-33 | Verify customer-only cardholder permission | Token permission assertion | 20 October rerun: PASS; customer permission present, operations permission absent | PASS |
| R17-34 | Hot-list active card as cardholder | POST /institutions/{inst}/cards/{id}/hotlist | 20 October rerun: HTTP 200; callerType CUSTOMER | PASS |
| R17-35 | Read cardholder-hotlisted card | GET /institutions/{inst}/cards/{id} | 20 October rerun: HTTP 200; BLOCKED | PASS |

# Phase 8: Card replacement

## W18 Replace a lost, stolen or compromised card

*Configure and activate the matching reason policy, then request replacement. These reasons must resolve to a new PAN. Verify the source and replacement lineage. If a request remains pending approval, record its state and any controlled refusal returned by approval/completion probes. Confirm the accepted PAN, PIN, control and account-link strategies against the allowed values.*

**Result: PASS — 30 of 30 checks passed**

The recorded LOST, STOLEN and COMPROMISED replacements resolved to a new PAN and blocked the source immediately. Baseline approve/reject/complete probes returned controlled HTTP 422 responses; successful approval/completion workflows are not established by these checks.

Baseline setup (16 October): 3 cards issued and activated (plan steps 29.3, 29.4, 29.5; 18 calls, all passed; data in the companion file under set_up).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0038 | Create LOST replacement policy | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies | HTTP 201 | PASS |
| S-0039 | Create STOLEN replacement policy | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies | HTTP 201 | PASS |
| S-0040 | Create COMPROMISED replacement policy | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies | HTTP 201 | PASS |
| S-0416 | J03 Report the eligibility card LOST | POST /institutions/{inst}/cards/{id}/holds/place | HTTP 200 | PASS |
| S-0417 | J03 A permanently restricted PAN family cannot be renewed onto the same PAN | POST /institutions/{inst}/cards/{id}/replacement | HTTP 422 | PASS |
| S-0418 | J03 The lost card was not replaced | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0609 | Replacement policies list | GET /institutions/{inst}/products/{id}/versions/{id}/replacement-policies?page=1&limit=20 | HTTP 200 | PASS |
| S-0610 | Replacement policy by id | GET /institutions/{inst}/products/{id}/versions/{id}/replacement-policies/{id} | HTTP 200 | PASS |
| S-0730 | Record LOST source identity | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0732 | Submit replacement: LOST new-PAN security replacement | POST /institutions/{inst}/cards/{id}/replacement | HTTP 202 | PASS |
| S-0734 | Read replacement request: LOST new-PAN security replacement | GET /institutions/{inst}/cards/{id}/replacement/{id} | HTTP 200 | PASS |
| S-0735 | Require completed replacement: LOST new-PAN security replacement | script check | Successor 01a14395-09ba-771c-bcce-f950c056b4c9; PAN strategy GENERATE_NEW_PAN | PASS |
| S-0736 | LOST source is BLOCKED immediately | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0737 | LOST successor has a new PAN family | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0745 | Record STOLEN source identity | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0747 | Submit replacement: STOLEN new-PAN security replacement | POST /institutions/{inst}/cards/{id}/replacement | HTTP 202 | PASS |
| S-0749 | Read replacement request: STOLEN new-PAN security replacement | GET /institutions/{inst}/cards/{id}/replacement/{id} | HTTP 200 | PASS |
| S-0750 | Require completed replacement: STOLEN new-PAN security replacement | script check | Successor 01a14395-31bf-7b6b-a3f3-423cb22e0cc3; PAN strategy GENERATE_NEW_PAN | PASS |
| S-0751 | STOLEN source is BLOCKED immediately | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0752 | STOLEN successor has a new PAN family | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0760 | Record COMPROMISED source identity | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0762 | Submit replacement: COMPROMISED new-PAN security replacement | POST /institutions/{inst}/cards/{id}/replacement | HTTP 202 | PASS |
| S-0764 | Read replacement request: COMPROMISED new-PAN security replacement | GET /institutions/{inst}/cards/{id}/replacement/{id} | HTTP 200 | PASS |
| S-0765 | Require completed replacement: COMPROMISED new-PAN security replacement | script check | Successor 01a14395-59d1-7eea-9292-48f5ebe3db4a; PAN strategy GENERATE_NEW_PAN | PASS |
| S-0766 | COMPROMISED source is BLOCKED immediately | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0767 | COMPROMISED successor has a new PAN family | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0776 | NAME_CHANGE is refused by eligibility | POST /institutions/{inst}/cards/{id}/replacement | HTTP 422 | PASS |
| S-0777 | Baseline replacement approve probe returns controlled refusal | POST /institutions/{inst}/cards/{id}/replacement/{id}/actions/approve | HTTP 422 | PASS |
| S-0778 | Baseline replacement reject probe returns controlled refusal | POST /institutions/{inst}/cards/{id}/replacement/{id}/actions/reject | HTTP 422 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0779 | Baseline replacement complete probe returns controlled refusal | POST /institutions/{inst}/cards/{id}/replacement/{id}/actions/complete | HTTP 422 | PASS |

## W19 Replace a damaged or expired card

*Use a physical-capable card and the exact active reason policy. Request replacement with the allowed PAN, shipping and production choices. Verify the resolved PAN strategy, the old and new card states and the production linkage. The important check is at the Switch: confirm the old card stops authorising in the way the policy says it should, whether that is immediately, on activation of the new card, or on first use. If the request stops at pending approval, record that state and stop there.*

**Result: PARTIAL - all 44 listed checks passed, including the 20 October Switch rerun; old-card authorisation remains unverified due to the known test-bed limitation.**

The table combines the 16 October baseline with the successful 20 October rerun of S-0501 (19.13), which confirmed separate source and successor Switch profiles. Known test-bed limitation. The NPCI simulator uses a different transaction card, so old-card authorisation after replacement remains unverified.

Baseline setup (16 October): 6 cards issued and activated (plan steps 16.1, 16.2, 16.10, 16.11, 17.1, 17.2, 19.1, 19.4, 24.1, 24.2, 29.6, 29.6a; 30 calls, all passed; data in the companion file under set_up).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0036 | Create RENEWAL replacement policy | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies | HTTP 201 | PASS |
| S-0037 | Create DAMAGED replacement policy | POST /institutions/{inst}/products/{id}/versions/{id}/replacement-policies | HTTP 201 | PASS |
| S-0394 | J01 Record the source identity | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0395 | J06 Enable a non-default control on the source | PUT /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0397 | Submit replacement: J01 same-PAN replacement | POST /institutions/{inst}/cards/{id}/replacement | HTTP 202 | PASS |
| S-0399 | Read replacement request: J01 same-PAN replacement | GET /institutions/{inst}/cards/{id}/replacement/{id} | HTTP 200 | PASS |
| S-0400 | Require completed replacement: J01 same-PAN replacement | script check | Successor 01a14393-3786-780c-828f-8b1b05827a76; PAN strategy PRESERVE_EXISTING_PAN | PASS |
| S-0401 | J05 The successor preserves the PAN and the account linkage | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0402 | J06 The successor control profile is rebuilt, not inherited | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0403 | J04 The source card state after a same-PAN replacement | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0404 | J04 The renewal switched off the source card-not-present channels | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-0405 | J07 A source with an existing replacement request is refused | POST /institutions/{inst}/cards/{id}/replacement | HTTP 422 | PASS |
| S-0411 | J09 A disallowed replacement reason is refused | POST /institutions/{inst}/cards/{id}/replacement | HTTP 422 | PASS |
| S-0412 | J09 A replacement reason with no policy is refused | POST /institutions/{inst}/cards/{id}/replacement | HTTP 422 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0413 | J02 A damaged replacement of a virtual-only card is refused | POST /institutions/{inst}/cards/{id}/replacement | HTTP 422 | PASS |
| S-0414 | J09 The refused replacements left the card alone | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0415 | J07 No successor was created for the refused reasons | GET /institutions/{inst}/cards?accountReferenceId={id}&page=1&limit=10 | HTTP 200 | PASS |
| S-0429 | K09 Record generation one | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0431 | Submit replacement: K09 first renewal | POST /institutions/{inst}/cards/{id}/replacement | HTTP 202 | PASS |
| S-0433 | Read replacement request: K09 first renewal | GET /institutions/{inst}/cards/{id}/replacement/{id} | HTTP 200 | PASS |
| S-0434 | Require completed replacement: K09 first renewal | script check | Successor 01a14393-60ae-7479-9ccd-79521c55ccbd; PAN strategy PRESERVE_EXISTING_PAN | PASS |
| S-0435 | K01 Generation two keeps the PAN and moves the expiry out | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0436 | K04 Generation one is still usable under BLOCK_ON_EXPIRY | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0437 | Activate K09 generation two | POST /institutions/{inst}/cards/{id}/actions/activate | HTTP 200 | PASS |
| S-0439 | Submit replacement: K09 second renewal | POST /institutions/{inst}/cards/{id}/replacement | HTTP 202 | PASS |
| S-0441 | Read replacement request: K09 second renewal | GET /institutions/{inst}/cards/{id}/replacement/{id} | HTTP 200 | PASS |
| S-0442 | Require completed replacement: K09 second renewal | script check | Successor 01a14393-70c9-70b8-928d-bce5d4e25a0d; PAN strategy PRESERVE_EXISTING_PAN | PASS |
| S-0443 | K09 Three generations are distinct records on one PAN | script check | Generations 01a14393-5092-76e2-b7e4-4856eac9a9ca -> 01a14393-60ae-7479-9ccd-79521c55ccbd -> 01a14393-70c9-70b8-928d-bce5d4e25a0d | PASS |
| S-0444 | K01 Generation three extends the expiry again | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0445 | K05 Freeze generation three | POST /institutions/{inst}/cards/{id}/holds/place | HTTP 200 | PASS |
| S-0446 | K05 Generation one is unaffected by the freeze | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0447 | K07 Renewal of a frozen card follows a defined rule | POST /institutions/{inst}/cards/{id}/replacement | HTTP 422 | PASS |
| S-0448 | K07 The frozen source kept its restriction | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0497 | Submit replacement: Q08 same-PAN renewal | POST /institutions/{inst}/cards/{id}/replacement | HTTP 202 | PASS |
| S-0499 | Read replacement request: Q08 same-PAN renewal | GET /institutions/{inst}/cards/{id}/replacement/{id} | HTTP 200 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0500 | Require completed replacement: Q08 same-PAN renewal | script check | Successor 01a14393-c277-70f4-b698-30de68049c17; PAN strategy PRESERVE_EXISTING_PAN | PASS |
| S-0501 | Q08 The Switch keeps the two same-PAN generations apart | script check | 20 October 2026 rerun (19.13): source and successor hold separate Switch profiles. | PASS |
| S-0553 | Submit replacement: T02 lineage renewal | POST /institutions/{inst}/cards/{id}/replacement | HTTP 202 | PASS |
| S-0554 | Read replacement request: T02 lineage renewal | GET /institutions/{inst}/cards/{id}/replacement/{id} | HTTP 200 | PASS |
| S-0555 | Require completed replacement: T02 lineage renewal | script check | Successor 01a14394-1599-7ea4-a206-9100cba946ed; PAN strategy PRESERVE_EXISTING_PAN | PASS |
| S-0556 | T02 The replacement record links both generations | GET /institutions/{inst}/cards/{id}/replacement/{id} | HTTP 200 | PASS |
| S-0557 | T03 The predecessor stays readable after renewal | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-0558 | T03 An obsolete replacement identifier is not found | GET /institutions/{inst}/cards/{id}/replacement/{id} | HTTP 404 | PASS |
| S-0775 | DAMAGED is refused by eligibility | POST /institutions/{inst}/cards/{id}/replacement | HTTP 422 | PASS |

## Phase 9: Product and configuration changes

### W20  Launch Product Version 2 with changed defaults

*Issue Card A under version 1. Create version 2, configure different defaults or policies, assign its profiles and design, activate it and issue Card B. Verify Card A stays linked to version 1 and Card B to version 2, and that Card A keeps the configuration it was issued under.*

**Result: PASS — 41 of 41 checks passed**

Version 1 is retired before version 2 activation because a product carries one ACTIVE version. Card A keeps every issuance fact from version 1 (41.55); Card B takes version 2 (41.43, 41.44).

Baseline preconditions (16 October): steps 41.07 to 41.07.8f (22 calls, all passed; data in the companion file).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0991 | Submit product (PENDING_APPROVAL) | POST /institutions/{inst}/products/{id}/actions/update | HTTP 200 | PASS |
| S-0992 | Activate product (ACTIVE) | POST /institutions/{inst}/products/{id}/actions/update | HTTP 200 | PASS |
| S-0993 | Submit BIN range assignment (PENDING_APPROVAL) | POST /institutions/{inst}/products/{id}/bin-range-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-0994 | Activate BIN range assignment (ACTIVE) | POST /institutions/{inst}/products/{id}/bin-range-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-0995 | Submit product version (PENDING_APPROVAL) | POST /institutions/{inst}/products/{id}/versions/{id}/actions/update | HTTP 200 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0996 | Activate product version (ACTIVE) | POST /institutions/{inst}/products/{id}/versions/{id}/actions/update | HTTP 200 | PASS |
| S-0998 | Submit issuance: CLIENT CARD A | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-1000 | Read issuance operation: CLIENT CARD A | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-1001 | Require issued card: CLIENT CARD A | script check | Card issued: 01a14395-d22c-790c-86ec-4af5a28e5be4 | PASS |
| S-1002 | W20 W21 W23 Retain Card A immutable issuance snapshot | script check | Recorded (see data file) | PASS |
| S-1003 | W20 Retain Card A controls before version change | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-1006 | W20 Create second version under same stable product | POST /institutions/{inst}/products/{id}/versions | HTTP 201 | PASS |
| S-1007 | W20 Set v2 physical-only form | PUT /institutions/{inst}/products/{id}/versions/{id}/card-form-policy | HTTP 200 | PASS |
| S-1008 | W20 Set changed v2 PIN retry policy | PUT /institutions/{inst}/products/{id}/versions/{id}/pin-policy | HTTP 200 | PASS |
| S-1009 | W20 Set v2 POS disabled by default | PUT /institutions/{inst}/products/{id}/versions/{id}/default-controls | HTTP 200 | PASS |
| S-1010 | W20 Set v2 DOMESTIC_ATM default | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-1011 | W20 Set v2 DOMESTIC_POS default | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-1012 | W20 Set v2 DOMESTIC_ECOMMERCE default | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-1013 | W20 Set v2 DOMESTIC_CONTACTLESS default | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-1014 | W20 Set v2 INTERNATIONAL_ATM default | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-1015 | W20 Set v2 INTERNATIONAL_POS default | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-1016 | W20 Set v2 INTERNATIONAL_ECOMMERCE default | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-1017 | W20 Set v2 INTERNATIONAL_CONTACTLESS default | POST /institutions/{inst}/products/{id}/versions/{id}/limits | HTTP 201 | PASS |
| S-1018 | W20 Set v2 personalisation policy | PUT /institutions/{inst}/products/{id}/versions/{id}/personalisation-policy | HTTP 200 | PASS |
| S-1021 | W20 Create independent v2 technology profile | POST /institutions/{inst}/products/{id}/versions/{id}/technology-profiles | HTTP 201 | PASS |
| S-1022 | W20 Activate v2 technology | POST /institutions/{inst}/products/{id}/versions/{id}/technology-profiles/{id}/actions/update | HTTP 200 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-1023 | W20 Create independent v2 CVM profile | POST /institutions/{inst}/products/{id}/versions/{id}/cvm-profiles | HTTP 201 | PASS |
| S-1024 | W20 Activate v2 CVM | POST /institutions/{inst}/products/{id}/versions/{id}/cvm-profiles/{id}/actions/update | HTTP 200 | PASS |
| S-1026 | W20 Retire own product v1 before v2 activation | POST /institutions/{inst}/products/{id}/versions/{id}/actions/update | HTTP 200 | PASS |
| S-1027 | Submit product version (PENDING_APPROVAL) | POST /institutions/{inst}/products/{id}/versions/{id}/actions/update | HTTP 200 | PASS |
| S-1028 | Activate product version (ACTIVE) | POST /institutions/{inst}/products/{id}/versions/{id}/actions/update | HTTP 200 | PASS |
| S-1029 | W20 Reject active version default rewrite | PUT /institutions/{inst}/products/{id}/versions/{id}/default-controls | HTTP 400 | PASS |
| S-1039 | Submit issuance: CLIENT CARD B | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-1041 | Read issuance operation: CLIENT CARD B | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-1042 | Require issued card: CLIENT CARD B | script check | Card issued: 01a14395-e462-7b9e-827a-c7f800eb13da | PASS |
| S-1043 | W20 W22 Card B selects v2 and expanded range | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-1044 | W20 Card B receives new POS default | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-1046 | W20 W21 W23 Card B new profile, design and CVK lineage | script check | Recorded (see data file) | PASS |
| S-1082 | W20 W21 W22 W23 Card A immutable facts survive all changes | script check | Recorded (see data file) | PASS |
| S-1083 | W20 W24 Card A controls unchanged after both products issued | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-1084 | W24 Product v2 defaults unchanged by other product | GET /institutions/{inst}/products/{id}/versions/{id}/default-controls | HTTP 200 | PASS |

## W21 Launch a new card-design version

*Issue one card with Design A. Create and activate Design B, fully configure a new product version, assign Design B, activate that version and issue another physical card. Earlier configuration must stay unchanged.*

**Result: PASS — 11 of 11 checks passed**

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0965 | W21 Create card design version 1 | POST /institutions/{inst}/card-designs | HTTP 201 | PASS |
| S-0966 | W21 Activate card design version 1 | POST /institutions/{inst}/card-designs/{id}/actions/update | HTTP 200 | PASS |
| S-1002 | W20 W21 W23 Retain Card A immutable issuance snapshot | script check | Recorded (see data file) | PASS |
| S-1004 | W21 Create card design version 2 | POST /institutions/{inst}/card-designs | HTTP 201 | PASS |
| S-1005 | W21 Activate card design version 2 | POST /institutions/{inst}/card-designs/{id}/actions/update | HTTP 200 | PASS |
| S-1019 | W21 Assign design version 2 only to product v2 | POST /institutions/{inst}/products/{id}/versions/{id}/card-design-assignments | HTTP 201 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-1020 | W21 Activate design v2 assignment | POST /institutions/{inst}/products/{id}/versions/{id}/card-design-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-1030 | W21 Reject active design artwork metadata rewrite | POST /institutions/{inst}/card-designs/{id}/actions/update | HTTP 400 | PASS |
| S-1046 | W20 W21 W23 Card B new profile, design and CVK lineage | script check | Recorded (see data file) | PASS |
| S-1082 | W20 W21 W22 W23 Card A immutable facts survive all changes | script check | Recorded (see data file) | PASS |
| S-1085 | W21 Original design version remains ACTIVE | GET /institutions/{inst}/card-designs/{id} | HTTP 200 | PASS |

## W22 Expand the BIN range available to a product

*Add a non-overlapping range under the existing BIN, activate it, assign it to the product and issue using the new range. Verify the new card reports the new range while earlier cards keep their original range.*

**Result: PASS — 12 of 12 checks passed**

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-1032 | W22 Create additional range on same BIN | POST /institutions/{inst}/bins/{id}/ranges | HTTP 201 | PASS |
| S-1033 | Submit BIN range (PENDING_APPROVAL) | POST /institutions/{inst}/bins/{id}/ranges/{id}/actions/update | HTTP 200 | PASS |
| S-1034 | Activate BIN range (ACTIVE) | POST /institutions/{inst}/bins/{id}/ranges/{id}/actions/update | HTTP 200 | PASS |
| S-1035 | W22 Assign expanded range to existing product | POST /institutions/{inst}/products/{id}/bin-range-assignments | HTTP 201 | PASS |
| S-1036 | Submit BIN range assignment (PENDING_APPROVAL) | POST /institutions/{inst}/products/{id}/bin-range-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-1037 | Activate BIN range assignment (ACTIVE) | POST /institutions/{inst}/products/{id}/bin-range-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-1039 | Submit issuance: CLIENT CARD B | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-1041 | Read issuance operation: CLIENT CARD B | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-1042 | Require issued card: CLIENT CARD B | script check | Card issued: 01a14395-e462-7b9e-827a-c7f800eb13da | PASS |
| S-1043 | W20 W22 Card B selects v2 and expanded range | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-1045 | W22 Expanded range records allocation and remaining capacity | GET /institutions/{inst}/bins/{id}/ranges/{id} | HTTP 200 | PASS |
| S-1082 | W20 W21 W22 W23 Card A immutable facts survive all changes | script check | Recorded (see data file) | PASS |

## W23 Rotate an HSM key used by a BIN

*Activate a protected successor key and its BIN assignment. Verify the prior key is retained as DECRYPT_ONLY and the prior assignment as VERIFY_ONLY. Confirm new issuance uses the successor and historical card metadata retains its original key reference. Historical-card transaction authorisation requires separate simulator evidence.*

**Result: PARTIAL - all 11 listed checks passed; F-01 closed on 21 October. Old-card authorisation remains unverified due to the known test-bed limitation.**

S-1094 to S-1097 use the successful 21 October rerun; other rows retain baseline evidence. Known test-bed limitation: the NPCI simulator uses a different transaction card, so old-card authorisation after rotation remains unverified.

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-0896 | Rotate an ACTIVE key | POST /institutions/{inst}/hsm-keys/{id}/rotations | HTTP 201 | PASS |
| S-1087 | W23 Record the prior BIN CVK before rotation | GET /institutions/{inst}/hsm-keys/{id} | HTTP 200 | PASS |
| S-1088 | W23 Generate real successor BIN CVK using HSM | POST /institutions/{inst}/hsm-keys | HTTP 201 | PASS |
| S-1090 | W23 Register rotated CVK version | POST /institutions/{inst}/hsm-keys/{id}/rotations | HTTP 201 | PASS |
| S-1091 | W23 Activate successor CVK | PUT /institutions/{inst}/hsm-keys/{id}/status | HTTP 200 | PASS |
| S-1092 | W23 Prior CVK is retained as DECRYPT_ONLY after successor activation | GET /institutions/{inst}/hsm-keys/{id} | HTTP 200 | PASS |
| S-1093 | W23 Create successor BIN key assignment | POST /institutions/{inst}/bins/{id}/hsm-key-assignments | HTTP 201 | PASS |
| S-1094 | W23 Activate successor BIN key assignment | PUT /institutions/{inst}/bins/{id}/hsm-key-assignments/{id}/status | 21 October rerun: HTTP 200; successor ACTIVE | PASS |
| S-1095 | W23 Verify old key assignment VERIFY_ONLY and successor ACTIVE | GET /institutions/{inst}/bins/{id}/hsm-key-assignments?keyPurpose=CVV_GENERATION&page=1&limit=100 | 21 October rerun: HTTP 200; old assignment VERIFY_ONLY, successor ACTIVE; exactly one ACTIVE assignment | PASS |
| S-1096 | W23 Postrotation issuance uses the successor CVK | POST /institutions/{inst}/cards; key-lineage check | 21 October rerun: issuance CONFIRMED; new card references the successor CVK and intended product version | PASS |
| S-1097 | W23 Historical card retains its original key reference | script check | 21 October rerun: historical card snapshot unchanged; original CVK reference retained | PASS |

## W24 Run multiple products on one BIN

*Create Product A and Product B under the same institution and BIN, preferably with separate non-overlapping BIN ranges. Configure and activate each independently, issue one card from each and verify the product, version and range lineage on both.*

**Result: PASS — 13 of 13 checks passed**

Baseline preconditions (16 October): steps 41.47 to 41.47.8f (20 calls, all passed; data in the companion file).

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-1068 | Submit product (PENDING_APPROVAL) | POST /institutions/{inst}/products/{id}/actions/update | HTTP 200 | PASS |
| S-1069 | Activate product (ACTIVE) | POST /institutions/{inst}/products/{id}/actions/update | HTTP 200 | PASS |

| Check | Test | Call | Observed | Result |
|---|---|---|---|---|
| S-1070 | Submit product version (PENDING_APPROVAL) | POST /institutions/{inst}/products/{id}/versions/{id}/actions/update | HTTP 200 | PASS |
| S-1071 | Activate product version (ACTIVE) | POST /institutions/{inst}/products/{id}/versions/{id}/actions/update | HTTP 200 | PASS |
| S-1072 | Submit BIN range assignment (PENDING_APPROVAL) | POST /institutions/{inst}/products/{id}/bin-range-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-1073 | Activate BIN range assignment (ACTIVE) | POST /institutions/{inst}/products/{id}/bin-range-assignments/{id}/actions/update | HTTP 200 | PASS |
| S-1075 | Submit issuance: CLIENT CARD C | POST /institutions/{inst}/cards | HTTP 202 | PASS |
| S-1078 | Read issuance operation: CLIENT CARD C | GET /institutions/{inst}/cards/issuance-operations/{id} | HTTP 200 | PASS |
| S-1079 | Require issued card: CLIENT CARD C | script check | Card issued: 01a14395-f5a3-7950-b9db-70ba1d7e5956 | PASS |
| S-1080 | W24 Independent product uses same BIN and its own version | GET /institutions/{inst}/cards/{id} | HTTP 200 | PASS |
| S-1081 | W24 Independent product retains its POS default | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-1083 | W20 W24 Card A controls unchanged after both products issued | GET /institutions/{inst}/cards/{id}/controls | HTTP 200 | PASS |
| S-1084 | W24 Product v2 defaults unchanged by other product | GET /institutions/{inst}/products/{id}/versions/{id}/default-controls | HTTP 200 | PASS |

## Phase 10: Full lifecycle

### W25 Run the complete lifecycle smoke test

*Run one fresh ordered lifecycle: institution, policies, HSM and data keys, BIN, range and key assignment, product, version, policies, profiles and design, activation, issuance, control change, limit change, PIN operation, replacement or hot-listing, and one authorisation. Read state after every write. This becomes the regression smoke test for future releases and major changes.*

**Result: PARTIAL - 57 baseline checks passed, with later hot-list confirmation added below. Successful PIN operations and final authorisation remain unverified.**

The lifecycle map combines baseline checks with the dated positive hot-list results. It is not evidence of a newly executed complete lifecycle. The PIN leg covers validation only; final authorisation is blocked by the known simulator card mismatch.

| Lifecycle leg | Checks | Check IDs | Result |
|---|---|---|---|
| Institution and policies | 01.2, 01.3, 32.7, 32.8 | S-0007, S-0008, S-0849, S-0850 | PASS |
| HSM and institution data keys | 00.4, 05.1, 05.2, 05.3, 05.4 | S-0005, S-0155, S-0156, S-0157, S-0158 | PASS |
| BIN, range and BIN key assignment | 01.4, 01.5, 01.6, 01.7, 05.9, 05.10, 05.11, 05.12 | S-0009, S-0010, S-0011, S-0012, S-0163, S-0164, S-0165, S-0166 | PASS |
| Product, version, policies, profiles and design | 02.1, 02.2, 02.3, 02.4, 02.5, 02.7p, 02.8e, 02.8f, 01.15, 01.16 | S-0023, S-0024, S-0025, S-0026, S-0027, S-0042, S-0046, S-0047, S-0020, S-0021 | PASS |
| Activation of assignments, product and version | 02.8, 02.8a, 02.8b, 02.9, 02.10, 02.11, 02.12, 02.13, 02.14 | S-0043, S-0044, S-0045, S-0048, S-0049, S-0050, S-0051, S-0052, S-0053 | PASS |
| Issuance and read-back | 10.6.1, 10.6.3, 10.8 | S-0272, S-0276, S-0278 | PASS |
| Card activation | 11.7, 11.8 | S-0301, S-0302 | PASS |

| Lifecycle leg | Checks | Check IDs | Result |
|---|---|---|---|
| Control change (read, change, read) | 14.3, 14.6, 14.9 | S-0356, S-0366, S-0370 | PASS |
| Limit read-back | 15.3 | S-0384 | PASS |
| PIN operation (validation paths) | 30.3, 30.4, 30.7 | S-0789, S-0791, S-0794 | PASS |
| Replacement (same-PAN renewal) and lineage | 16.5.1, 16.5.3, 16.6, 24.7 | S-0397, S-0400, S-0401, S-0556 | PASS |
| Hot-list (operations and cardholder) | 31.28, 31.29, 31.34, 31.35 (20 October) | R17-28, R17-29, R17-34, R17-35 | PASS |
| Authorisation | — | Known test-bed limitation: NPCI simulator uses a different card | NOT EXECUTED |
| Final listings and direct reads | 24.3, 24.4, 24.5, 24.10 | S-0549, S-0550, S-0551, S-0559 | PASS |

# 5 Additional CMS checks

The plan also exercises CMS behaviour beyond Table 3. This summary uses 16 October baseline evidence, with check 19.10 updated from the successful 20 October rerun (section 3.2). Original data remains under supplementary in the companion file.

| Area | Checks | Passed | Failed | Not executed | Plan steps |
|---|---|---|---|---|---|
| Card activation with an existing hold | 6 | 6 | 0 | 0 | 13.1 to 13.4 |
| Card closure from every state | 26 | 26 | 0 | 0 | 18.1 to 18.15 |
| CMS to Switch synchronisation view | 4 | 4 | 0 | 0 | 19.2 to 19.10 |
| Idempotency on holds and configuration writes | 9 | 9 | 0 | 0 | 20.1 to 20.5, 35.3a to 35.3b |
| Authentication and tenancy boundaries | 10 | 9 | 0 | 1 | 23.1 to 23.11, 39.1 to 39.2 |
| Card reads and operation visibility | 4 | 4 | 0 | 0 | 24.3 to 24.10 |
| Independent holds and cross-card isolation | 17 | 17 | 0 | 0 | 25.1 to 25.13 |
| Canonical lifecycle commands and hold escalation | 22 | 22 | 0 | 0 | 31.3 to 31.22 |
| API envelope and request validation | 5 | 5 | 0 | 0 | 36.1 to 36.5 |
| Pagination behaviour | 11 | 11 | 0 | 0 | 37.1 to 37.13 |
| Outbox operations | 2 | 2 | 0 | 0 | 38.1 to 38.3 |

Tenant isolation remains unproven because the probe used a platform-scoped client. CMS-to-Switch comparison 19.10 passed in the 06:24 run. Closed-card and other later Switch wait timeouts remain recorded in sections 3.2 and 3.3; no successful rerun resolving those specific failures is supplied.

# Appendix A Companion data file

The companion JSON preserves baseline samples under run, sections and supplementary. Its current_assessment and the dated results files describe the consolidated report. Baseline failure statuses and settings are historical records; dated report rows identify their superseding evidence.

| Path | Content |
|---|---|
| document | Current report metadata; baseline document metadata is archived separately |
| run | Historical 16 October run identifier, timing, settings and totals |
| fixtures | Identifiers of the synthetic test data created for the run |
| sections.W01 … sections.W25 | Historical per-scenario baseline checks and fixtures; later results are identified under current_assessment and results/ |

| Path | Content |
|---|---|
| sections.Wnn.checks[] | Per check: check_id (S-nnnn), label, call (method, URL, headers, body), response (status, headers, body), assertions[], result, timing |
| supplementary | Historical additional CMS checks; section 5 identifies the later 19.10 override |
| findings | Historical baseline findings and supporting check IDs |
| redaction | What was redacted and how many values were affected |
| current_assessment | Current scenario status/counts, evidence sources, R-reference mapping and unresolved integration/coverage items |
| results/*.json | Dated Switch-sync, PVK/ZPK, hot-list, W15 limit and W23 rotation verification; baseline responses remain archived |

The companion JSON remains the original combined evidence file. Its npci_simulator_run, historical_switch_examples and sections.W26 to sections.W43 entries support the separate Switch report.

# Appendix B  Package contents

| File | Purpose |
|---|---|
| Falcon_CMS_Scenario_Test_Report.pdf / .docx | This report |
| Falcon_CMS_Test_Data_and_Evidence.json | Companion data file (see Appendix A) |
| jmeter/CMS_V2_LIFECYCLE_FUNCTIONAL.jmx | Original 16 October baseline plan; later focused/extended runs are documented separately |
| jmeter/runtime.properties.example | Historical baseline runtime-property example with secrets removed |
| results/samples.csv | One row per executed sample: time, check ID, plan step, label, HTTP status, result, role (CHECK, SET_UP, PRECONDITION, FIXTURE, POLL, HARNESS, INFORMATIONAL, GROUP_TOTAL), scenario reported under, duration |
| README.txt, SHA256SUMS.txt | Package guide and checksums |
| results/report-current-assessment-2026-10-21.json | Current consolidated assessment and report-reference index |
| results/*-verification-*.json | Dated verification summaries used by current scenario rows |